Name: Sanyukta Futane

Email: sanyuktafutane01@gmail.com

Task-2:

Goal

Demonstrate data analysis skills using Pandas.

Requirements

- Load and inspect a CSV dataset
- Clean missing or incorrect data
- Apply filtering, grouping, and aggregation
- Explain insights in simple words

Deliverables

1. Notebook or Python script
2. Short insight summary

OUTPUT SCREENSHOT

Data.csv:

data.csv

```
ID,Name,Age,Department,Salary,Experience,City
1,Alice,25,IT,50000,2,Nagpur
2,Bob,30,HR,45000,5,Mumbai
3,Charlie,28,Finance,60000,4,Pune
4,David,35,IT,70000,8,Delhi
5,Eve,32,HR,48000,6,Bangalore
6,Frank,29,Finance,52000,3,Hyderabad
7,Grace,27,IT,51000,3,Nagpur
8,Hank,31,HR,46000,5,Mumbai
9,Ivy,26,Finance,58000,2,Pune
10,Jack,34,IT,72000,9,Delhi
11,Kate,29,HR,47000,4,Bangalore
12,Leo,33,Finance,61000,7,Hyderabad
13,Mia,24,IT,49000,1,Nagpur
14,Nina,36,HR,50000,10,Mumbai
15,Oscar,38,Finance,65000,12,Pune
16,Paul,27,IT,53000,3,Delhi
17,Quinn,31,HR,45500,5,Bangalore
18,Rita,28,Finance,59000,4,Hyderabad
19,Sam,35,IT,71000,8,Nagpur
20,Tina,30,HR,46000,6,Mumbai
21,Uma,26,Finance,57000,2,Pune
22,Victor,34,IT,73000,9,Delhi
23,Wendy,29,HR,48000,4,Bangalore
24,Xavier,32,Finance,62000,6,Hyderabad
25,Yara,25,IT,50000,2,Nagpur
26,Zack,37,HR,51000,11,Mumbai
27,Aman,28,Finance,60000,4,Pune
28,Bina,33,IT,68000,7,Delhi
29,Chetan,31,HR,47000,5,Bangalore
30,Diya,27,Finance,58000,3,Hyderabad
31,Elena,29,IT,52000,3,Nagpur
32,Fahad,35,HR,49500,8,Mumbai
33,Gauri,26,Finance,56000,2,Pune
34,Harsh,38,IT,75000,12,Delhi
```

data.csv

```
29,Chetan,31,HR,47000,5,Bangalore
30,Diya,27,Finance,58000,3,Hyderabad
31,Elena,29,IT,52000,3,Nagpur
32,Fahad,35,HR,49500,8,Mumbai
33,Gauri,26,Finance,56000,2,Pune
34,Harsh,38,IT,75000,12,Delhi
35,Ishan,30,HR,46000,6,Bangalore
36,Jyoti,28,Finance,59000,4,Hyderabad
37,Kunal,24,IT,48000,1,Nagpur
38,Lata,36,HR,50500,10,Mumbai
39,Manish,39,Finance,67000,13,Pune
40,Neha,27,IT,54000,3,Delhi
41,Om,31,HR,47000,5,Bangalore
42,Pooja,29,Finance,60000,4,Hyderabad
43,Rahul,34,IT,72000,9,Nagpur
44,Sneha,30,HR,46000,6,Mumbai
45,Tarun,26,Finance,57000,2,Pune
46,Usha,33,IT,69000,7,Delhi
47,Vikas,28,HR,45500,4,Bangalore
48,Anita,35,Finance,64000,8,Hyderabad
49,Rohit,27,IT,53000,3,Nagpur
50,Kiran,32,HR,48000,6,Mumbai
51,Alice,25,IT,50000,2,Nagpur
52,,30,HR,45000,5,Mumbai
53,Charlie,,Finance,60000,4,Pune
54,David,35,IT,,8,Delhi
55,Eve,32,HR,48000,,Bangalore
56,Frank,29,Finance,52000,3,
57,Grace,27,IT,51000,3,Nagpur
58,Hank,31,HR,46000,5,Mumbai
59,Ivy,26,Finance,58000,2,Pune
60,Jack,34,IT,72000,9,Delhi
```

Output:

```
PS C:\Users\lenovo\OneDrive\Desktop\sanyukta\Alfido-Tech-Internship> & C:/Users/lenovo/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/lenovo/OneDrive/Desktop/sanyukta/Alfido-Tech-Internship/task2.py
First 5 rows:
   ID     Name   Age Department   Salary  Experience       City
0   1    Alice  25.0         IT  50000.0         2.0     Nagpur
1   2      Bob  30.0         HR  45000.0         5.0     Mumbai
2   3  Charlie  28.0    Finance  60000.0         4.0       Pune
3   4    David  35.0         IT  70000.0         8.0      Delhi
4   5      Eve  32.0         HR  48000.0         6.0  Bangalore

Dataset Info:
<class 'pandas.DataFrame'>
RangeIndex: 60 entries, 0 to 59
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   ID          60 non-null     int64
 1   Name        59 non-null     str
 2   Age         59 non-null     float64
 3   Department  60 non-null     str
 4   Salary      59 non-null     float64
 5   Experience  59 non-null     float64
 6   City        59 non-null     str
dtypes: float64(3), int64(1), str(3)
memory usage: 3.4 KB
None

Summary Statistics:
              ID        Age        Salary  Experience
count  60.000000  59.000000     59.000000   59.000000
mean   30.500000  30.440678  55474.576271    5.372881
std    17.464249   3.856348   8803.126271    2.982121
min     1.000000  24.000000  45000.000000    1.000000
```

```
Summary Statistics:
              ID        Age        Salary  Experience
count  60.000000  59.000000     59.000000   59.000000
mean   30.500000  30.440678  55474.576271    5.372881
std    17.464249   3.856348   8803.126271    2.982121
min     1.000000  24.000000  45000.000000    1.000000
25%    15.750000  27.000000  48000.000000    3.000000
50%    30.500000  30.000000  52000.000000    5.000000
75%    45.250000  33.500000  60000.000000    7.500000
max    60.000000  39.000000  75000.000000   13.000000

After Cleaning:
<class 'pandas.DataFrame'>
Index: 55 entries, 0 to 59
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   ID          55 non-null     int64
 1   Name        55 non-null     str
 2   Age         55 non-null     float64
 3   Department  55 non-null     str
 4   Salary      55 non-null     float64
 5   Experience  55 non-null     float64
 6   City        55 non-null     str
dtypes: float64(3), int64(1), str(3)
memory usage: 3.4 KB
None

Skipping filtering (column not found)

Skipping grouping (columns not found)

Cleaned data saved as 'cleaned_data.csv'
```

Cleaned data set:

cleaned_data.csv

```
ID,Name,Age,Department,Salary,Experience,City
1,Alice,25.0,IT,50000.0,2.0,Nagpur
2,Bob,30.0,HR,45000.0,5.0,Mumbai
3,Charlie,28.0,Finance,60000.0,4.0,Pune
4,David,35.0,IT,70000.0,8.0,Delhi
5,Eve,32.0,HR,48000.0,6.0,Bangalore
6,Frank,29.0,Finance,52000.0,3.0,Hyderabad
7,Grace,27.0,IT,51000.0,3.0,Nagpur
8,Hank,31.0,HR,46000.0,5.0,Mumbai
9,Ivy,26.0,Finance,58000.0,2.0,Pune
10,Jack,34.0,IT,72000.0,9.0,Delhi
11,Kate,29.0,HR,47000.0,4.0,Bangalore
12,Leo,33.0,Finance,61000.0,7.0,Hyderabad
13,Mia,24.0,IT,49000.0,1.0,Nagpur
14,Nina,36.0,HR,50000.0,10.0,Mumbai
15,Oscar,38.0,Finance,65000.0,12.0,Pune
16,Paul,27.0,IT,53000.0,3.0,Delhi
17,Quinn,31.0,HR,45500.0,5.0,Bangalore
18,Rita,28.0,Finance,59000.0,4.0,Hyderabad
19,Sam,35.0,IT,71000.0,8.0,Nagpur
20,Tina,30.0,HR,46000.0,6.0,Mumbai
21,Uma,26.0,Finance,57000.0,2.0,Pune
22,Victor,34.0,IT,73000.0,9.0,Delhi
23,Wendy,29.0,HR,48000.0,4.0,Bangalore
24,Xavier,32.0,Finance,62000.0,6.0,Hyderabad
25,Yara,25.0,IT,50000.0,2.0,Nagpur
26,Zack,37.0,HR,51000.0,11.0,Mumbai
27,Aman,28.0,Finance,60000.0,4.0,Pune
28,Bina,33.0,IT,68000.0,7.0,Delhi
29,Chetan,31.0,HR,47000.0,5.0,Bangalore
30,Diya,27.0,Finance,58000.0,3.0,Hyderabad
31,Elena,29.0,IT,52000.0,3.0,Nagpur
32,Fahad,35.0,HR,49500.0,8.0,Mumbai
33,Gauri,26.0,Finance,56000.0,2.0,Pune
34,Harsh,38.0,IT,75000.0,12.0,Delhi
```

cleaned_data.csv

```
30,Diya,27.0,Finance,58000.0,3.0,Hyderabad
31,Elena,29.0,IT,52000.0,3.0,Nagpur
32,Fahad,35.0,HR,49500.0,8.0,Mumbai
33,Gauri,26.0,Finance,56000.0,2.0,Pune
34,Harsh,38.0,IT,75000.0,12.0,Delhi
35,Ishan,30.0,HR,46000.0,6.0,Bangalore
36,Jyoti,28.0,Finance,59000.0,4.0,Hyderabad
37,Kunal,24.0,IT,48000.0,1.0,Nagpur
38,Lata,36.0,HR,50500.0,10.0,Mumbai
39,Manish,39.0,Finance,67000.0,13.0,Pune
40,Neha,27.0,IT,54000.0,3.0,Delhi
41,Om,31.0,HR,47000.0,5.0,Bangalore
42,Pooja,29.0,Finance,60000.0,4.0,Hyderabad
43,Rahul,34.0,IT,72000.0,9.0,Nagpur
44,Sneha,30.0,HR,46000.0,6.0,Mumbai
45,Tarun,26.0,Finance,57000.0,2.0,Pune
46,Usha,33.0,IT,69000.0,7.0,Delhi
47,Vikas,28.0,HR,45500.0,4.0,Bangalore
48,Anita,35.0,Finance,64000.0,8.0,Hyderabad
49,Rohit,27.0,IT,53000.0,3.0,Nagpur
50,Kiran,32.0,HR,48000.0,6.0,Mumbai
51,Alice,25.0,IT,50000.0,2.0,Nagpur
57,Grace,27.0,IT,51000.0,3.0,Nagpur
58,Hank,31.0,HR,46000.0,5.0,Mumbai
59,Ivy,26.0,Finance,58000.0,2.0,Pune
60,Jack,34.0,IT,72000.0,9.0,Delhi
```

CODE:

```
import pandas as pd

# 1. Load CSV dataset
df = pd.read_csv('data.csv')

# 2. Inspect the dataset
print("First 5 rows:")
print(df.head())

print("\nDataset Info:")
print(df.info())

print("\nSummary Statistics:")
print(df.describe())

# 3. Data Cleaning

# Handle missing values
df_cleaned = df.dropna()

# Remove duplicates
df_cleaned = df_cleaned.drop_duplicates()

print("\nAfter Cleaning:")
print(df_cleaned.info())

# 4. Filtering (Example: filter rows where a column value > some value)
# Replace 'column_name' with actual column
if 'column_name' in df_cleaned.columns:
    filtered_data = df_cleaned[df_cleaned['column_name'] > df_cleaned['column_name'].mean()]
    print("\nFiltered Data:")
    print(filtered_data.head())
```

```
else:
    print("\nSkipping filtering (column not found)")

# 5. Grouping and Aggregation
# Replace 'category_column' and 'value_column' with actual names
if 'category_column' in df_cleaned.columns and 'value_column' in df_cleaned.columns:
    grouped = df_cleaned.groupby('category_column')['value_column'].mean()
    print("\nGrouped Data (Average values):")
    print(grouped)
else:
    print("\nSkipping grouping (columns not found)")

# 6. Save cleaned dataset
df_cleaned.to_csv('cleaned_data.csv', index=False)

print("\nCleaned data saved as 'cleaned_data.csv'")
```

- **GITHUB LINK:**
  ➔ https://github.com/Sanyukta-Futane01/Alfido-Tech-Internship